

An Internship Report
Of
AI-Based Hiring Prediction System

Submitted

To

School of Computer Science and Engineering

ILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B-Tech (CSE)



By

Roll Number of Student:- 2410031087

Name of Student: ASIM AZHAR

Batch: 2026-27

CANDIDATE'S DECLARATION

I, ASIM AZHAR, do hereby declare that the internship report titled “**AI-Based Hiring Prediction System**” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

Student Name: ASIM AZHAR

Roll No.: 2410031087

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work. I express my gratitude to my parents for their blessings.

Signature

Student Name: ASIM AZHAR

Roll No.: 2410031087

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INTERNSHIP COMPLETION CERTIFICATE

LINUXWORLD INFORMATICS PVT LTD

#1 Awarded Summer Internship Industrial Training Program in India & Asia

CERTIFICATE OF TRAINING

This is to certify that

Asim Azhar

has successfully completed
Research Based Summer Industrial Internship training on

Transforming Business Intelligence with Data Science,
Machine Learning & Artificial Intelligence for Predictive Analytics,
Intelligent Decision-Making, Customer Behavior Analysis,
Advanced Data Modeling, AI-Driven Automation & Real-Time
Insights – **Unleashing the Power of Data Science,
Machine Learning and Artificial Intelligence for
Next-Generation Innovation and Business Success.**



Certificate No. : LWIPL-2026-1285
Duration : 27th June – 31st July, 2026

Authorized Signatory
LinuxWorld Informatics Pvt. Ltd.



Internship Offer Letter



LinuxWorld Informatics Pvt. Ltd.

CIN: U72900RJ2012PTC039171

Ref : LWIPL-JPR- 2026-1142

Date: 8th May, 2026

To,
The Training & Placement Officer
IILM University, Greater Noida

This is to certify that **Mr. Asim Azhar** Student of **IILM University, Greater Noida** will be undergoing the Summer Internship Program on "**Data Science with Machine Learning & AI**" at LinuxWorld Informatics Pvt Ltd, under the mentorship of The World Record Holder & CTO - **Mr. Vimal Daga**, for a period of 45 Days as per the official college Curriculum.

The Management will view his performance continuously during his summer period of research and development.

Regards,
LinuxWorld Informatics Pvt Ltd


Ms. Preeti Daga



+91 9829105960



lwindia.com
training@lwindia.com



Plot No. 5, Krishna Tower, Next to Triveni Nagar Flyover,
Gopalpura Bypass, Jaipur, Rajasthan - 302015

PROJECT DESCRIPTION

4.1 Introduction

During the Summer Internship at **LinuxWorld Informatics Pvt. Ltd.**, a project titled **“AI-Based Hiring Prediction System”** was developed as part of the internship project work. The project focused on applying Data Science, Machine Learning, and Artificial Intelligence techniques to the recruitment and Human Resource Management domain.

The AI-Based Hiring Prediction System is a machine learning application designed to assist recruiters in making hiring decisions based on a candidate's professional and academic information. The system considers factors such as **skills, certifications, education, work experience, number of projects, salary expectations, and AI test results** to predict whether a candidate is likely to be hired.

The project demonstrates how machine learning and text-processing techniques can be combined to analyze candidate information and provide a predicted hiring decision along with the probability of selection.

4.2 Organization Profile

LinuxWorld Informatics Pvt. Ltd. is the organization where the Summer Internship project was undertaken. According to the internship certificate, the project was completed by **Mr. Asim Azhar from IILM University, Greater Noida**, under the mentorship of **Mr. Syed Jibbran Ali, Sr. Technical Consultant**. The internship/project duration mentioned in the certificate was **27th June 2026 to 31st July 2026**.

The certificate identifies LinuxWorld Informatics Pvt. Ltd. as the organization under which the project work and report were completed. It also carries the organization's reference number **LW/JPR/2026/1285, dated 31st July 2026**.

4.3 Problem Statement

Recruitment involves evaluating candidates using multiple academic and professional factors such as education, skills, certifications, experience, projects, salary expectations, and assessment results. Manually analyzing these factors for a large number of candidates can require significant time and may result in inconsistent decision-making.

The problem addressed by this project is to develop a **machine learning-based system that can analyze candidate information and predict whether a candidate is likely to be hired**.

The proposed system processes both numerical and textual candidate information. Textual fields such as **skills, certifications, and job roles** are converted into numerical representations using **TF-IDF**, while numerical features are scaled before being provided to machine learning models.

The system is intended to support recruiters by providing a faster and more consistent approach to initial candidate screening and hiring prediction.

4.4 Problem Objectives

The major objectives of the AI-Based Hiring Prediction System are:

- To develop a machine learning system capable of predicting hiring decisions based on candidate information.
- To preprocess and prepare hiring-related datasets for machine learning.
- To analyze candidate information including skills, certifications, education, experience, projects, salary requirements, and AI test results.
- To convert categorical and textual information into machine-readable numerical representations.
- To use **TF-IDF** for processing textual fields such as skills, certifications, and job roles.
- To train and compare multiple machine learning algorithms, including **Logistic Regression, Random Forest, Support Vector Machine (SVM), and K-Nearest Neighbors (KNN)**.
- To evaluate the performance of the trained models using accuracy and other performance measures.
- To optimize the selected model using **GridSearchCV**.
- To provide recruiters with a predicted hiring decision and the probability of a candidate being selected.
- To demonstrate the application of machine learning in Human Resource Management and recruitment.

4.5 Scope of the Project

The scope of the project is focused on using machine learning and data processing techniques to assist with candidate screening and hiring prediction.

The system covers the following areas:

- Loading and processing a hiring dataset.
- Exploration and visualization of candidate data.
- Removal of unnecessary data columns.
- Encoding of categorical variables.

- Processing textual information such as skills, certifications, and job roles.
- Conversion of text into TF-IDF vectors.
- Scaling of numerical features.
- Training and testing of multiple machine learning models.
- Comparison of model performance.
- Optimization of the selected model using GridSearchCV.
- Prediction of hiring decisions for new candidates.

The certificate also identifies several possibilities for extending the system, including a **web-based recruiter interface, candidate ranking, automated interview scheduling, Explainable AI (XAI), cloud deployment using AWS or Azure, and integration with Applicant Tracking Systems (ATS).**

4.6 Technologies and Tools Used

The project was implemented using Python and common Data Science and Machine Learning tools.

Technology/Tool	Purpose
Python	Primary programming language used for the project
Google Colab / Jupyter Notebook	Development and execution environment
Pandas	Data handling and preprocessing
NumPy	Numerical operations and data processing
Matplotlib	Data visualization
Seaborn	Data visualization and exploratory analysis
Scikit-learn	Machine learning model development and evaluation
TF-IDF	Conversion of textual data into numerical vectors
StandardScaler	Scaling numerical features
Pipeline	Combining preprocessing and machine learning steps
GridSearchCV	Hyperparameter optimization and model improvement

The certificate specifically states that Pandas and NumPy were used for handling and preparing data, while Matplotlib and Seaborn were used for visualization. Scikit-learn was used for implementing Logistic Regression, Random Forest Classifier, SVM, and KNN. TF-IDF was used for processing textual information, while StandardScaler, Pipeline, and GridSearchCV supported preprocessing and model optimization.

4.7 System Architecture

The **AI-Based Hiring Prediction System** follows a machine learning pipeline in which candidate data is processed before being provided to multiple machine learning models.

System Workflow: Hiring Dataset → Data Preprocessing → Text Processing (TF-IDF) → Feature Scaling → Train-Test Split → Model Training → Model Comparison → GridSearchCV → Best Model → Hiring Prediction

This architecture is based on the working methodology and technologies explicitly described in the internship certificate.

Main Components of the system are:

- 1. Data Input:** The system starts by loading the hiring dataset containing academic, professional, and other candidate-related information.
- 2. Data Preprocessing:** Unnecessary columns are removed and categorical variables are converted into numerical values.
- 3. Text Processing:** Skills, certifications, and job roles are combined and converted into numerical vectors using TF-IDF.
- 4. Feature Scaling:** Numerical features are scaled using StandardScaler to improve the machine learning process.
- 5. Model Training:** Multiple machine learning algorithms are trained and evaluated.
- 6. Model Optimization:** GridSearchCV is used to optimize the selected model.
- 7. Prediction:** The final system accepts candidate information and produces a hiring decision along with the candidate's probability of being selected.

4.8 Methodology

The methodology followed during the development of the AI-Based Hiring Prediction System consisted of the following steps:

- 1. Load the Hiring Dataset:** The hiring dataset was loaded into the Python development environment for further analysis and processing.

2. Data Exploration and Visualization: The dataset was explored to understand its structure, features, and patterns. Visualization techniques were used to better understand the data.

3. Data Cleaning: Unnecessary columns were identified and removed from the dataset.

4. Encoding Categorical Variables: Categorical variables were converted into numerical representations so that they could be processed by machine learning algorithms.

5. Text Feature Processing: Important textual fields such as **Skills, Certifications, and Job Role** were combined for further processing.

6. TF-IDF Vectorization: TF-IDF was applied to convert the textual information into numerical vectors that could be used as machine learning features.

7. Feature Scaling: Numerical features were scaled using **StandardScaler**.

8. Train-Test Split: The processed dataset was divided into training and testing sets to train and evaluate the machine learning models.

9. Model Training: Multiple machine learning algorithms were trained:

- Logistic Regression
- Random Forest
- Support Vector Machine (SVM)
- K-Nearest Neighbors (KNN)

10. Model Comparison: The performance of the different models was compared using accuracy and other performance measures.

11. Model Optimization: The best-performing model was further optimized using **GridSearchCV**.

12. Hiring Prediction: The final system can accept information about a new candidate and provide a predicted hiring decision along with the probability of selection.

4.9 Expected Outcomes

The expected outcome of the project is an **AI-Based Hiring Prediction System capable of supporting recruiters in candidate screening and hiring decisions.**

The system is expected to:

- Predict whether a candidate is likely to be hired.
- Process both numerical and textual candidate information.

- Combine information such as education, skills, certifications, and experience.
- Reduce the time required for initial candidate screening.
- Provide more consistent support for hiring decisions.
- Provide a hiring prediction along with the probability of candidate selection.
- Demonstrate the practical application of machine learning in Human Resource Management.

According to the certificate, the project demonstrates that machine learning can help make candidate screening easier and decision-making faster while providing a practical application of AI and machine learning in the recruitment domain.

The project can be further extended through:

- Integration with real-time recruitment portals.
- Resume parsing using Natural Language Processing (NLP).
- Use of deep learning models for improved prediction accuracy.
- Development of a web-based interface for recruiters.
- Automated interview scheduling.
- Candidate ranking.
- Explainable AI (XAI) for greater prediction transparency.
- Cloud deployment using AWS or Azure.
- Integration with Applicant Tracking Systems (ATS).

4.10 Certificates of completion and other communication mail proof must be attached

The internship confirmation was formally communicated by LinuxWorld Informatics Pvt. Ltd. through email on 8 May 2026. The email contained the confirmation letter for the selected Summer Project and stated that the letter could also be submitted to the college as proof of the internship.



Asim Azhar <asimazhar0905@gmail.com>

Summer Intern at LinuxWorld Informatics Pvt. Ltd. under Mr. Vimal Daga

1 message

Arti - LinuxWorld Summer Team <arti.bansal@lwindia.com>
To: Asim Azhar <asimazhar0905@gmail.com>

Fri, May 8, 2026 at 6:01 PM

Dear Learner,

Greetings from **LinuxWorld!**

Please find attached the Confirmation Letter for your Summer project selected. The same can be submitted to your college as well.

under the mentorship of The World Record Holder, **Mr. Vimal Daga**

- **Linkedin Profile**- <https://www.linkedin.com/in/vimaldaga/>
- **Insta Profile** - <https://www.instagram.com/vimaldaga.india/?hl=en>

Time to showcase & let your network with recruiters know that soon you will be **READY WITH MARKET DEMANDED SKILLS & LIVE PROJECT** which will make you ready for your placement in respective Domain

Don't forget to post your CONFIRMATION Letter & SUMMER BADGE on Social media handles & use the following hashtags 🥳

#RightEducation #VimalDaga #Linuxworld #summerindustrialtraining #btechsummertraining #SummerInternship

All the other details will soon be reaching your inbox about the timings & other related aspects. Stay tuned..

To ensure a seamless learning experience, **we kindly request you to acknowledge the receipt of this letter.**

Thank you for choosing LinuxWorld for your Summer Program. We look forward to supporting you throughout your journey.

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Best Regards,

Co-ordinator Linux World

Co-ordinator - LinuxWorld

On the Vision of Awakening the youth through a culture of Right Education

<https://www.lwindia.com/>

arti.bansal@lwindia.com

M: +91-9351009002 - Arti Bansal



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